AI-ML Internship IBM SkillsBuild

Presented By: Code Breakers 2.O

Group Members:

1. **Nishant**
2. **Sahil**
3. **Punit Verma**
4. **Karamveer**
5. **Maneesha**
6. **Seema**

College: Guru Jambheshwar University of Science and Technology, Hisar

Title:

AI-Powered Career Guidance Chatbot Aligned with UNSDG 4 – Quality Education

Introduction:

Career guidance plays a vital role in shaping the future of students and professionals. With the rapid technological advancements and evolving job market, there is a growing need for accurate and accessible career advice. This project presents an AI-powered chatbot designed to provide personalized career guidance, aligning with the United Nations Sustainable Development Goal (UNSDG) 4: Quality Education. The chatbot aims to assist users in making informed decisions about their future, whether it involves further education or stepping into the job market.

Problem Statement:

Many students and young professionals struggle to identify the right career path due to a lack of accessible, personalized, and unbiased guidance. Traditional career counseling methods are limited by human availability, cost, and geographic constraints. Additionally, the wide variety of career options often confuses individuals who do not have sufficient support systems or mentors. There is an urgent need for a scalable and intelligent solution that can provide timely and relevant career advice.

Objective:

The primary objective of this project is to develop an AI-based chatbot that offers real-time, personalized career guidance to individuals based on their interests, academic background, and goals. The chatbot aims to:

- Democratize access to quality career guidance.
- Support both academic and job-related decision-making.
- Encourage continuous learning and skill development.
- Contribute to achieving UNSDG 4 by ensuring inclusive and equitable quality education.

Why This Problem:

India, like many other countries, faces challenges in career planning, especially in rural and underserved areas. A large number of students are unaware of their potential career options due to the absence of proper mentorship and guidance infrastructure. Even urban students often rely on inconsistent sources of information. This leads to poor career choices, dissatisfaction, and underemployment. By addressing this gap, we can empower individuals to make informed choices and align their goals with evolving industry demands, thus contributing to sustainable development and societal progress.

Solution:

Overview:

The proposed solution is an AI-powered chatbot that provides intelligent and interactive career guidance. It is designed to simulate a human-like conversation while understanding user queries and responding with relevant suggestions. Whether the user is looking to pursue higher education or is interested in exploring job opportunities, the chatbot serves as a virtual counselor, available 24/7.

Features:

- **Personalized Guidance**: Tailored responses based on user's academic background, interests, and aspirations.
- **Education Pathways**: Suggests suitable courses, degrees, or certifications.
- **Job Recommendations**: Offers suggestions for jobs based on skillset and industry trends.

- **User-Friendly Interface**: Accessible via a simple web interface.
- **Continuous Learning**: The chatbot evolves and improves with more interactions using OpenAI's natural language model.
- **Goal Alignment**: Encourages users to align their plans with long-term professional and educational goals.

Technical Implementation:

- **Programming Language**: The core of the chatbot is developed using **Python** due to its simplicity and the availability of powerful AI libraries.
- **Platform Used**:
  - **Google Colab**: For training and prototyping AI models efficiently using cloud GPU resources.
  - **Visual Studio Code (VS Code)**: For developing and debugging the main application.
- **Libraries and Frameworks**:
  - **OpenAI API**: For natural language understanding and generation.
  - **Flask**: A lightweight web framework used to connect the backend with the frontend and handle HTTP requests.
- **Web Technologies**:
  - **HTML & CSS**: Used for building the user interface of the chatbot. The front-end is clean, responsive, and easy to use for all age groups.
- **Deployment**:
  - The chatbot is hosted locally or on a web server, where users can access it through a web page. The backend processes the input, interacts with the AI model, and returns meaningful responses.
- **Security and Privacy**:
  - Basic input validation and sanitization are implemented to protect against misuse or injection attacks.
  - User data is kept confidential and not stored unless explicitly allowed.

Conclusion:

The Career Guidance Chatbot is an innovative step toward making career counseling accessible, affordable, and inclusive. It not only addresses the problem of limited guidance resources but also aligns with the global objective of Quality Education (UNSDG 4). Through AI, we can ensure that every individual receives equal opportunity to explore, plan, and pursue a career that aligns with their skills and passion.